NeuroSense: Automated Muscle Fasciculation Monitoring Using GMM-Based Foreground Segmentation

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Summary

The detection of moving regions in image sequences is a fundamental step in many vision systems, including automated visual surveillance, human-machine interface, and very low-bandwidth telecommunications. We initially developed the system in MATLAB using Gaussian mixture models for fasciculation detection [1]. Although effective, MATLAB has limitations in expanding and testing advanced methods. Python, with its rich set of machine learning and computer vision libraries, offers a better platform for future development.

We have successfully migrated our code to Python using OpenCV BackgroundSubtractorMOG2 [2], which is also GMM-based. With Python, NeuroSense can become a smarter and more accurate tool, helping clinicians identify fasciculations earlier.

Statement of need

Early detection is critical in conditions such as motor neuron disease (MND), where timely intervention can significantly impact patient outcomes. The transition of Gaussian Mixture Model (GMM) implementation from MATLAB to Python addresses the need for an open-source, scalable, and extensible platform to support advanced statistical modeling and image analysis tasks.

Key Needs Driving the Transition:

- Scalability: With tools like OpenCV, sci-kit-learn, and distributed computing frameworks, Python provides a robust platform for handling large-scale datasets and computationally intensive tasks.
- Customization and Flexibility: The OpenCV GMM implementation allows for detailed customization of parameters and models, addressing domain-specific requirements beyond the constraints of the built-in functions of MATLAB.
- Community Support: Python's global community fosters rapid development, shared resources, and continuous updates, ensuring long-term sustainability and innovation.

Background

Motor neuron disease (MND) is characterized by progressive degeneration of motor neurons leading to muscle weakness and wasting and reduced mobility, speech, swallowing, and respiratory abilities [1,3]. These diseases are currently incurable [1,3,4]. To facilitate trials of new therapeutic interventions, there is a significant need to identify sensitive markers of neuromuscular degeneration to support early diagnosis and provide sensitive outcome measures of progression [3]. Ultrasound imaging provides a non-invasive [5], real-time view of larger muscle areas, making it an alternative for fasciculation detection. We introduce a novel application of adaptive foreground detection using Gaussian Mixture Models to identify fasciculations in ultrasound image sequences (see Figure 1). The tool was tested in five skeletal muscles in healthy individuals and individuals affected by MND.

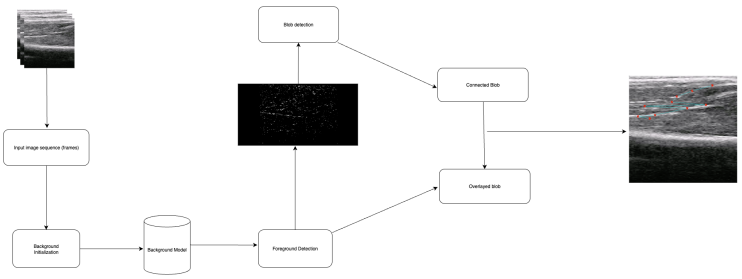


Figure 1: Low-level Architecture

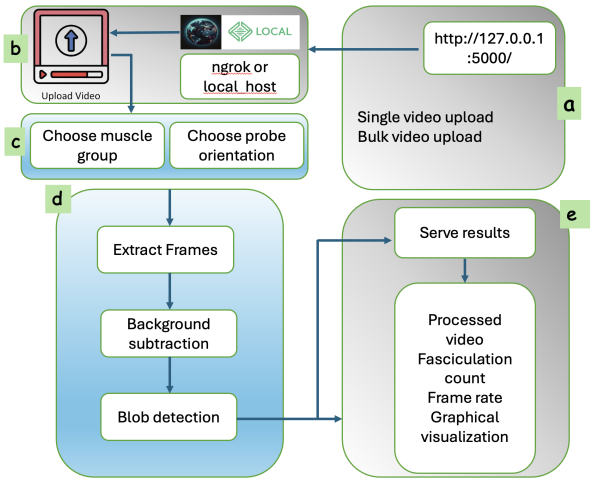


Figure 2: High-level Architecture

The NeuroSense Tool

NeuroSense is an open-source software tool developed to objectively detect and quantify muscle fasciculations in ultrasound video sequences. It uses an adaptive background subtraction method based on Gaussian Mixture Models [6], allowing it to identify involuntary twitch events that are often subtle and brief. The tool processes ultrasound videos by modeling pixel-level motion over time, effectively distinguishing background motion from muscle activity associated with fasciculations. It was designed to operate on a range of skeletal muscles and to be resistant to variations in ultrasound probe orientation, an important factor in clinical workflows.

Software Availability

The NeuroSense software is openly available:

- GitHub: https://github.com/Dheeraj791/NeuroSense
- Zenodo DOI: 10.5281/zenodo.17138130

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Citation

If you use NeuroSense in your research, please cite:

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